

No. of Functional Features Included in the Solution

Date	24 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Dataset Loading	Load historical flood prediction dataset using Pandas	Data Collection	Done	High
2	Exploratory Data Analysis (EDA)	Perform descriptive statistics, distribution plots, box plots, and correlation heatmap	Data Analysis	Done	1. High
3	Data Preprocessing	Handle missing values, detect outliers using IQR, split dataset, and apply StandardScaler	Data Preprocessing	Done	High
4	Machine Learning Model Training	Train Decision Tree, Random Forest, KNN, and XGBoost classifiers	Machine Learning	Done	High
5	Model Evaluation & Comparison	Compare models using Accuracy Score, Confusion Matrix, and Classification Report	Model Evaluation	Done	High
6	Model Saving	Save the best-performing model (<code>floods.save</code>) and scaler (<code>scaler.save</code>) using Joblib	Model Deployment	Done	Medium
7	Flask Web Application	Develop a web application for flood prediction using Flask	Backend Development	Done	High
8	User Prediction Interface	Create HTML/CSS pages for Home, Prediction, and Result with instant flood prediction	Frontend Development	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	8
Additional / Nice-to-Have Features	0
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Prediction Form, Result Page
2	Backend / Logic	4	Flask Application, Input Processing, Model Loading, Prediction Logic
3	Database / Storage	2	Flood Dataset, Saved Model (<code>floods.save</code>) and Scaler (<code>scaler.save</code>)
4	API / Integration	0	Not implemented (Future Scope: Weather API Integration)
5	Security / Authentication	0	Not required for the current project